The documentation and process conversion measures necessary to comply with this document shall be completed by 4 Novemberr 2006.

INCH-POUND

MIL-PRF-19500/336J
4 August 2006
SUPERSEDING
MIL-PRF-19500/336H
21 July 2005

PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, UNITIZED, DUAL-TRANSISTOR, PNP, SILICON, TYPES 2N3810, 2N3810L, 2N3810U, 2N3811, 2N3811L, AND 2N3811U, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for two electrically isolated, matched PNP, silicon transistors as one dual unit. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (similar to TO-78), figures 2 and 3 for unencapsulated devices, and figure 4 for U devices.

* 1.3 Maximum ratings, unless otherwise specified, $T_C$ =+25°C.

| Type | $I_C$ | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ |
|---|---|---|---|---|
| | mA dc | V dc | V dc | V dc |
| All types | 50 | 60 | 60 | 5 |

| $P_T$ (1) $T_A$ = +25°C | | $P_T$ (2) $T_C$ = +25°C | | $R_{\theta JA}$ | | $R_{\theta JC}$ | | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|
| One section | Both sections | One section | Both sections | One section | Both sections | One section | Both sections | |
| mW | mW | mW | mW | °C/W | °C/W | °C/W | °C/W | °C |
| 200 | 350 | 300 | 450 | 875 | 500 | 583 | 350 | -65 to +200 |

(1) For $T_A$ > +25°C, derate linearly 1.143 mW/°C, one section; 2.000 mW/°C, both sections. See figure 5 and 6.
(2) For $T_C$ > +25°C, derate linearly 1.714 mW/°C, one section; 2.571 mW/°C, both sections.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .



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1.4 Primary electrical characteristics at $T_A$ = +25°C, pulsed (see 4.5.2).

| Limit | $h_{FE3}$ through $h_{FE4}$ | | $\lvert h_{fe} \rvert_2$ | $V_{BE(sat)\,2}$ | $V_{CE(sat)\,2}$ |
|---|---|---|---|---|---|
| | $V_{CE}$ = 5 V dc; $I_C$ = 100 μA dc Through $I_C$ = 1 mA dc | | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc; f = 100 MHz | $I_C$ = 1 mA dc; $I_B$ = 100 μA dc | $I_C$ = 1 mA dc; $I_B$ = 100 μA dc |
| | 2N3810 2N3810L | 2N3811 2N3811L | | V dc | V dc |
| Minimum | 150 | 300 | 1 | | |
| Maximum | 450 | 900 | 5 | 0.8 | 0.25 |

1.5 Primary electrical matching characteristics of each individual section.

| Limit | $\frac{h_{FE3-1}}{h_{FE3-2}}$ $V_{CE}$ = 5 V dc; $I_C$ = 100 μA dc (1) | $\lvert V_{BE1} - V_{BE2} \rvert_2$ $V_{CE}$ = 5 V dc; $I_C$ = 100 μA dc | $\lvert \Delta (V_{BE1} - V_{BE2}) \Delta T_A \rvert 1$ $V_{CE}$ = 5 V dc $I_C$ = 100 μA dc $T_A$ = +25°C and –55°C | $\lvert \Delta (V_{BE1} - V_{BE2}) \Delta T_A \rvert 2$ $V_{CE}$ = 5 V dc; $I_C$ = 100 μA dc $T_A$ = +125°C and +25°C |
|---|---|---|---|---|
| | | mV dc | mV dc | mV dc |
| Minimum | 0.9 | | | |
| Maximum | 1.0 | 3 | 0.8 | 1.0 |

(1) The larger number will be placed in the denominator.



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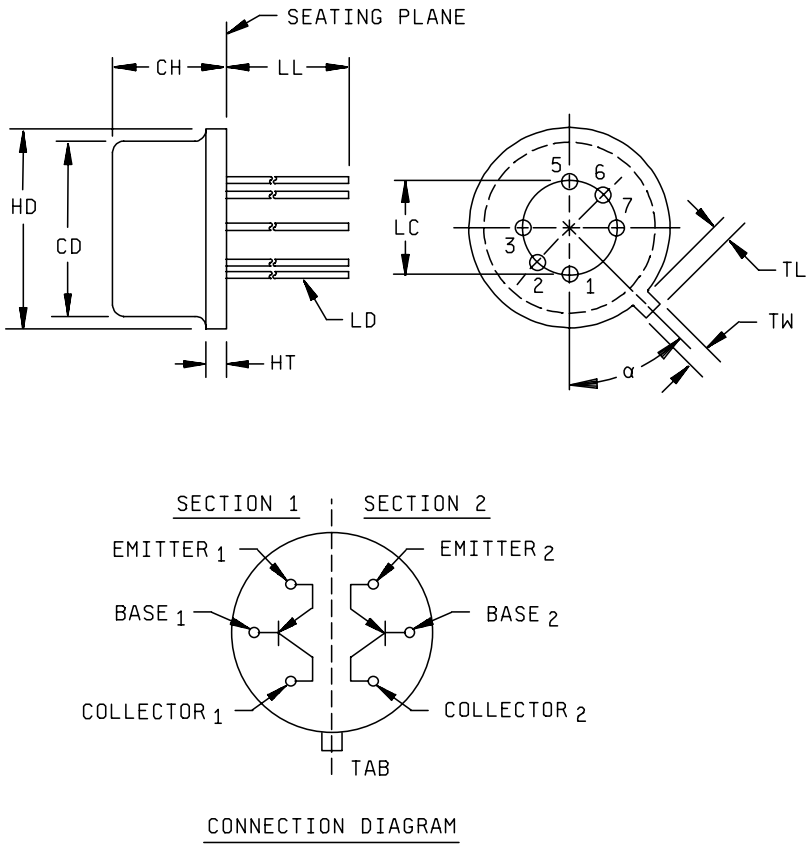



FIGURE 1. Physical dimensions (similar to TO-78).



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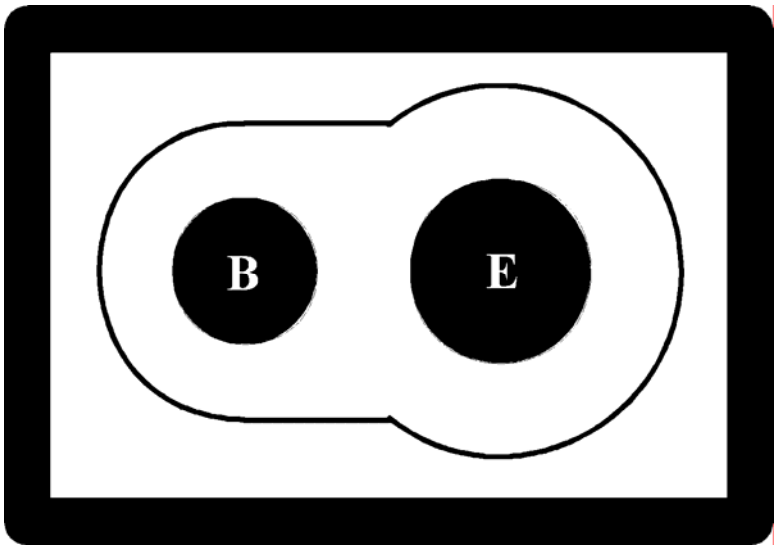
| LTR | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | |
| CH | .150 | .185 | 3.81 | 4.70 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| HT | .009 | .041 | 0.23 | 1.04 | |
| LC | .200 BSC | | 5.08 BSC | | |
| LD | .016 | .021 | 0.41 | 0.53 | 10 |
| LL | See notes 10, 11, and 13 | | | | |
| $\alpha$ | 45°TP | | 45°TP | | 9 |
| TL | .029 | .045 | 0.74 | 1.14 | 5, 6 |
| TW | .028 | .034 | 0.71 | 0.86 | 4, 5 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Refer to rules for dimensioning Semiconductor Product Outlines included in Publication No. 95.
4. Lead number 4 and 8 omitted on this variation.
5. TW must be held to a minimum length of .021 inch (0.53 mm).
6. LL measured from maximum HD.
7. Details of outline in this zone optional.
8. CD shall not vary more than .010 inch (0.25mm) in zone P. This zone is controlled for automatic handling.
9. Leads at gauge plane .054 - .055 inch (1.37 - 1.40 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at a maximum material condition (MMC) relative to the tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure described on gauge drawing GS-1.
10. LD applies to LL minimum.
11. r (radius) applies to both inside corners of tab.
12. For transistor types 2N3810 and 2N3811, LL is .500 inch (12.70 mm) minimum, and .750 inch (19.05 mm) maximum. (TO-99).
13. For transistor types 2N3810L and 2N3811L, LL is 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum.
14. In accordance with AMSE Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 1. Physical dimensions (similar to TO-78) – Continued.



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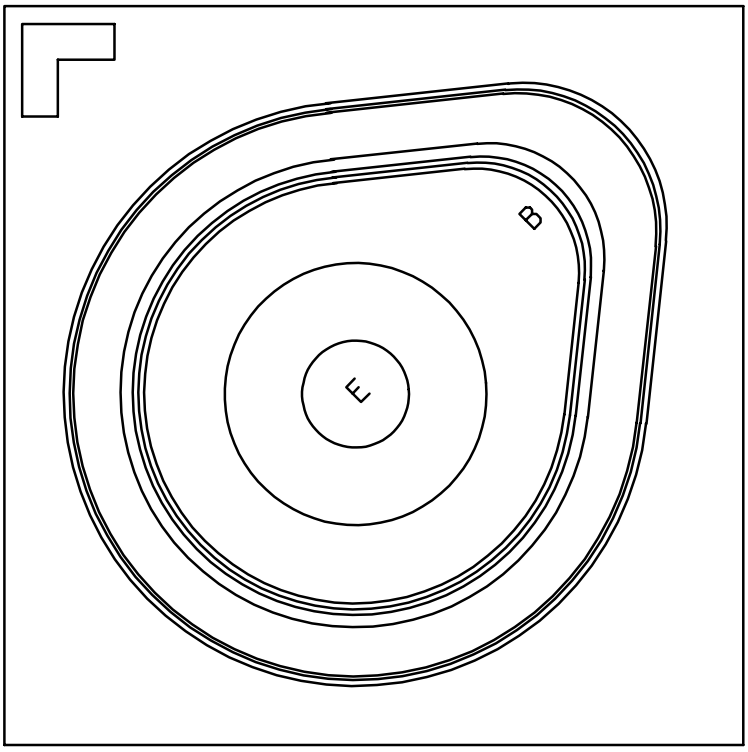



1. Chip size..... .015 x .019 inch ±.001 inch (0.38 x 0.48 mm ±0.025 mm).
2. Chip thickness..... .010 ±0.0015 inch (0.25 ±0.038 mm).
3. Top metal..... Aluminum 15,000Å minimum, 18,000Å nominal.
4. Back metal..... A. Gold 2,500Å minimum, 3,000Å nominal.
   B. Eutectic Mount - No Gold.
5. Backside..... Collector.
6. Bonding pad.....B = .003 inch (0.07 mm), E = .004 inch (0.10 mm) diameter.
7. Passivation…… ……$Si_3N_4$ (Silicon Nitride) 2kÅ minimum, 2.2kÅ nominal.

FIGURE 2. JANHC and JANKC A-version die dimensions.



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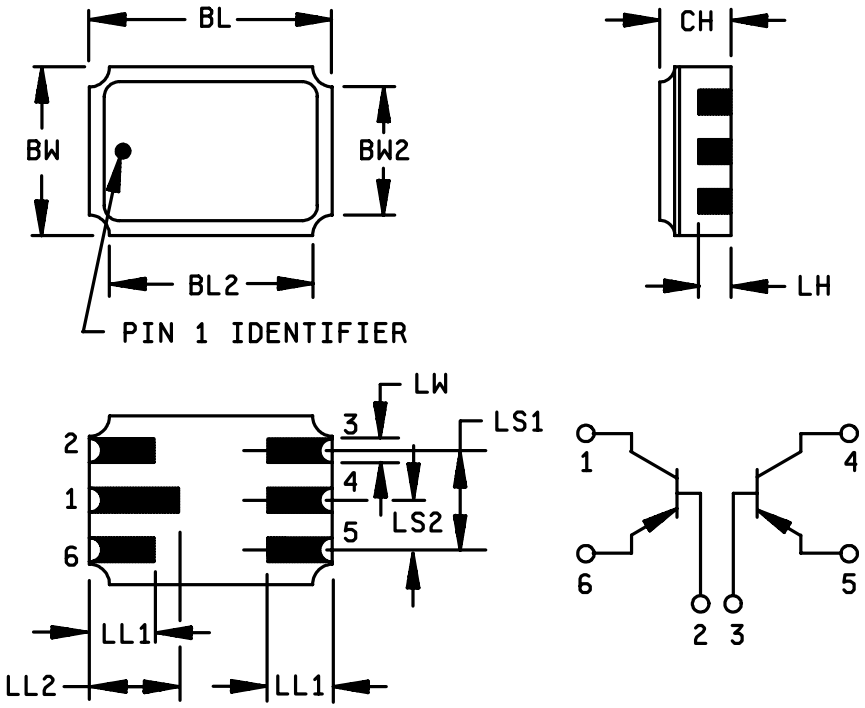



| | |
|---|---|
| Die size: | .018 x .018 inch (0.457 mm). |
| Die thickness: | .008 ±.0016 inch (0.20 ±0.041 mm). |
| Base pad: | .0025 inch diameter (0.064 mm). |
| Emitter pad: | .003 inch diameter (0.076 mm). |
| Back metal: | Gold, 6500 ±1950 Ang. |
| Top metal: | Aluminum, 19500 ±2500 Ang. |
| Back side: | Collector. |
| Glassivation: | $SiO_2$, 7500 ±1500 Ang. |

FIGURE 3. JANHC and JANKC B-version die dimensions.



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .240 | .250 | 6.10 | 6.35 |
| BL2 | | .250 | | 6.35 |
| BW | .165 | .175 | 4.19 | 4.45 |
| BW2 | | .175 | | 4.45 |
| CH | .044 | .080 | 1.12 | 2.03 |
| LH | .026 | .039 | 0.66 | 0.99 |
| LL1 | .060 | .070 | 1.52 | 1.78 |
| LL2 | .082 | .098 | 2.08 | 2.49 |
| LS1 | .095 | .105 | 2.41 | 2.67 |
| LS2 | .045 | .055 | 1.14 | 1.40 |
| LW | .022 | .028 | 0.56 | 0.71 |

| Pin no. | Transistor |
|---|---|
| 1 | Collector no. 1 |
| 2 | Base no. 1 |
| 3 | Base no. 2 |
| 4 | Collector no. 2 |
| 5 | Emitter no. 2 |
| 6 | Emitter no. 1 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with AMSE Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 4. Physical dimensions (2N3810U and 2N3811U).



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

| | |
|---|---|
| $\frac{h_{FE3-1}}{h_{FE3-2}}$ | Static forward-current-gain-ratio. The matching ratio of the static forward-current transfer ratios of each section. |
| $\lvert V_{BE1} - V_{BE2} \rvert$ | Absolute value of base-emitter-voltage differential between the individual sections. |
| $\lvert \Delta V_{BE1-2}\ (T1) - \Delta V_{BE1-2}\ (T2) \rvert$ | Absolute value of the algebraic difference between the base-emitter-voltage differentials between the individual sections at two different temperatures. |



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3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (similar to TO-78), figures 2 and 3 for unencapsulated devices, and figure 4 for U devices.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, 1.5, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2.1 Group E thermal response. With extremely small junction devices such as this one, a true thermal impedance cannot be measure, only calculated. While “thermal response” has been substituted for “thermal impedance” herein, the terms, units and procedure as essentially unchanged. Each supplier shall submit a thermal response ($Z_{\theta JX}$) histogram of the entire qualification lot. The histogram data shall be taken prior to the removal of devices that are atypical for thermal response. Thermal response curves (from $Z_{\theta JX}$ test pulse time to $R_{\theta JX}$ minimum steady-state time) of the best device in the qual lot and the worst device in the qual lot (that meets the supplier proposed screening limit), or from the thermal grouping, shall be submitted. The optimal test conditions and proposed initial thermal response screening limit shall be provided in the qualification report. Data indicating how the optimal test conditions were derived for $Z_{\theta JX}$ shall also be submitted. The proposed maximum thermal response $Z_{\theta JX}$ screening limit shall be submitted. The qualifying activity may approve a different $Z_{\theta JX}$ limit for conformance inspection end-point measurements as applicable. Equivalent data, procedures, or statistical process control plans may be used for part, or all, of the above requirements. The approved thermal response conditions and limit for $Z_{\theta JX}$ shall be used by the supplier in screening and table I, subgroup 2. The approved thermal resistance conditions for $R_{\theta JX}$ shall be used by the supplier for conformance inspection. For product families with similar thermal characteristics based on the same physical and thermal die, package, and construction combination (thermal grouping), the supplier may use the same thermal response curves.



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* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) 3c | Thermal response, method 3131 of MIL-STD-750. See 4.3.3. | Thermal response, method 3131 of MIL-STD-750. See 4.3.3. |
| 7 | Optional | Optional |
| 9 | $I_{CBO2}$, $h_{FE4}$ | Not applicable |
| 10 | 24 hours minimum | 24 hours minimum |
| 11 | $I_{CBO2}$; $h_{FE4}$;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater.<br>$\Delta h_{FE4}$ = ±15 percent. | $I_{CBO2}$ ,$h_{FE4}$ |
| 12 | See 4.3.2 | See 4.3.2 |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE4}$ = ±15 percent. | Subgroup 2 and the base emitter voltage (nonsaturated) (absolute value of differential-change with temperature) tests of subgroup 3 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE4}$ = ±15 percent. |
| 14 | Required | Required |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, “Discrete Semiconductor Die/Chip Lot Acceptance”. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

* 4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc. Power shall be applied to achieve $T_J$ =135°C minimum and using a minimum $P_D$ = 75 percent of $P_T$ maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site’s burn-in data and performance history will be essential criteria for burn-in modification approval.

* 4.3.3 Thermal response. For very small junction devices, such as this, the term “thermal response” shall be used in lieu of “thermal impedance” although measurements shall be performed the same way as thermal impedance in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$ ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See group E, subgroup 4 herein.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of subgroup 1 and 2, of table I herein, inspection only (table E-VIb, group B, subgroup 1 is not required to be performed since solderability and resistance to solvents testing is performed in A1 herein).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.3 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2 and 4.5.3 herein.

* 4.4.2.1 Group B inspection (JANS), table E-VIa of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = 10 V dc, 2,000 cycles, adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C. |
| B5 | 1027 | $V_{CB}$ = 10 V dc; $P_D \geq$ 100 percent of maximum rated $P_T$ (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)<br><br>Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table E-VIa, adjust $T_A$ or $P_D$ to achieve $T_J$ = +275°C minimum.<br><br>Option 2: 216 hours minimum, sample size = 45, c = 0; adjust $T_A$ or $P_D$ to achieve a $T_J$ = +225°C minimum. |

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = 10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |



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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.3 herein.

* 4.4.3.1 Group C inspection (JANS), table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; (not applicable for U devices). |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and 4.3.3 herein. |
| C6 | 1026 | 1,000 hours at $V_{CB}$ = 10 V dc; power shall be applied to achieve $T_J$ = +150°C minimum and a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3 n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |

* 4.4.3.2 Group C inspection (JAN, JANTX, and JANTXV), table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; not applicable for U device. |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and 4.3.3 herein. |
| C6 | | Not applicable. |

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein; delta measurements shall be in accordance with the applicable steps of 4.5.3.



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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Testing of units. All specified electrical tests, including electrical measurements (end-points) and delta requirement tests, shall be performed equally on both sections of the transistor types covered herein, except where the electrical characteristic being evaluated applies to the transistor as a device entity.

4.5.2 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.3 Delta requirements. Delta requirements shall be as specified below:

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | Unit |
|---|---|---|---|---|---|---|
| | | Method | Conditions | | | |
| 1 | Collector-base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 50 V dc | $\Delta I_{CB02}$ (1) | 100 percent of initial value or 8 nA dc, whichever is greater. | |
| 2 | Forward current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc; pulsed see 4.5.2 | $\Delta h_{FE4}$ (1) | ±25 percent change from initial reading. | |

(1) Devices which exceed the table I limits for this test shall not be accepted.

4.5.4 Disposition of leads when testing characteristics of each section. During the measurement of the characteristic of each section, the leads of the section not under test shall be open-circuited.

4.5.5 Forward-current-gain ratio. The value for the forward-current-gain ratio for each individual section of a dual unit shall be measured using method 3076 of MIL-STD-750. The forward-current-gain ratio shall be calculated by dividing one of the values by the other. If possible, this ratio shall be measured directly to improve accuracy.

4.5.6 Base-emitter-voltage differential. The base-emitter-voltage differential shall be determined by connecting the emitters of the individual sections together, applying specified electrical test conditions to each individual section in accordance with method 3066 of MIL-STD-750, test condition B and measuring the absolute value of the voltage between the bases of the individual sections of a dual unit.

4.5.7 Base-emitter-voltage differential change with temperature. The value of the base-emitter-voltage differential shall be measured at the two specified temperatures in accordance with 4.5.6 except that the identities of the individual sections shall be maintained. The absolute value of the algebraic difference between the values at the two temperature extremes shall be calculated. A mathematical formula for this parameter is:

$$\left| (V_{BE1} - V_{BE2})\, T1 \; - \; (V_{BE1} - V_{BE2})\, T2 \right|$$

4.5.8 Noise figure test. Noise figure shall be measured using a model No. 2173C/2181 Quan Tech Laboratories test set, or equivalent. Conditions shall be as specified in table I.

4.5.9 Noise figure (wideband) test. Wideband noise figure shall be measured using a model No. 512 Quan Tech Laboratories test set, or equivalent. Conditions shall be as specified in table I.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical examination 3/ | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ 4/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ 5/ | 1022 | n = 15 devices, c = 0 | | | | |
| Temp cycling 3/ 4/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Hermetic seal 4/ 6/<br>Fine leak<br>Gross leak | 1071 | n = 22 devices, c = 0 | | | | |
| Electrical measurements 4/ | | Table I, subgroup 2 | | | | |
| Bond strength 3/ 4/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hours or $T_A$ = +300°C at t = 2 hours n = 11 wires, c = 0 | | | | |
| Decap internal visual (design verification) 4/ | 2075 | n = 4 devices, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal response | 3131 | See 4.3.3. | $Z_{\theta JX}$ | | | °C/W |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 60 V dc (see 4.5.2) | $I_{CBO1}$ | | 10 | µA dc |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D; $I_C$ = 100 µA dc; pulsed (see 4.5.2) | $V_{(BR)CEO}$ | 60 | | V dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 5 V dc | $I_{EBO1}$ | | 10 | µA dc |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 50 V dc | $I_{CBO2}$ | | 10 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 4 V dc | $I_{EBO2}$ | | 10 | nA dc |
| Forward-current transfer ratio 2N3811, 2N3811L, U only | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 1 µA dc | $h_{FE1}$ | 75 | | |

See footnotes at end of table.



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* TABLE I. Group A inspection – Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Forward-current transfer ratio<br>2N3810, 2N3810L, U<br>2N3811, 2N3811L, U | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 10 μA dc | $h_{FE2}$ | <br>100<br>225 | | |
| Forward-current transfer ratio<br>2N3810, 2N3810L, U<br>2N3811, 2N3811L, U | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 100 μA dc | $h_{FE3}$ | <br>150<br>300 | <br>450<br>900 | |
| Forward-current transfer ratio<br>2N3810, 2N3810L, U<br>2N3811, 2N3811L, U | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc | $h_{FE4}$ | <br>150<br>300 | <br>450<br>900 | |
| Forward-current transfer ratio<br>2N3810, 2N3810L, U<br>2N3811, 2N3811L, U | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 10 mA dc | $h_{FE5}$ | <br>125<br>250 | | |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 100 μA dc, $I_B$ = 10 μA dc | $V_{CE(sat)1}$ | | 0.2 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 1 mA dc, $I_B$ = 100 μA dc | $V_{CE(sat)2}$ | | 0.25 | V dc |
| Base-emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 100 μA dc, $I_B$ = 10 μA dc | $V_{BE(sat)1}$ | | 0.7 | V dc |
| Base-emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 1 mA dc, $I_B$ = 100 μA dc | $V_{BE(sat)2}$ | | 0.8 | V dc |
| Base-emitter voltage (nonsaturated) | 3066 | Test condition B; $V_{CE}$ = 5 V dc; $I_C$ = 100 μA dc | $V_{BE(ON)}$ | | 0.7 | V dc |
| Forward-current transfer ratio (gain ratio) | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 100 μA dc (see 4.5.5) | $h_{FE3-1}$ / $h_{FE3-2}$ | 0.9 | 1.0 | |
| Base emitter voltage (nonsaturated) (absolute value of differential) 7/ | 3066 | Test condition B<br>$V_{CE}$ = 5 V dc; $I_C$ = 10 μA dc (see 4.5.6) | $\lvert V_{BE1} - V_{BE2} \rvert 1$ | | 5 | mV dc |
| Base emitter voltage (nonsaturated) (absolute value of differential) 7/ | 3066 | Test condition B<br>$V_{CE}$ = 5 V dc; $I_C$ = 100 μA dc (see 4.5.6) | $\lvert V_{BE1} - V_{BE2} \rvert 2$ | | 3 | mV dc |

See footnotes at end of table.



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* TABLE I. Group A inspection – Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Base emitter voltage (nonsaturated) (absolute value of differential) 7/ | 3066 | Test condition B $V_{CE}$ = 5 V dc; $I_C$ = 10 mA dc (see 4.5.6) | $\lvert V_{BE1} - V_{BE2} \rvert$3 | | 5 | mV dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Base emitter voltage (absolute value of differential-change with temperature) | 3066 | Test condition B $V_{CE}$ = 5 V dc; $I_C$ = 100 µA dc $T_A$ = +125°C and +25°C (see 4.5.7) | $\lvert \Delta V_{BE1} - V_{BE2} \Delta T_A \rvert$2 | | 1.0 | mV dc |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 50 V dc | $I_{CBO3}$ | | 10 | µA dc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 100 µA dc | $h_{FE6}$ | | | |
| 2N3810, 2N3810L, U<br>2N3811, 2N3811L, U | | | | 60<br>100 | | |
| Base emitter voltage (absolute value of differential-change with temperature) | 3066 | Test condition B $V_{CE}$ = 5 V dc; $I_C$ = 100 µA dc $T_A$ = +25°C and -55°C (see 4.5.7) | $\lvert \Delta V_{BE1} - V_{BE2} \Delta T_A \rvert$1 | | 0.8 | mV dc |
| Subgroup 4 | | | | | | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 5 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 5 | pF |
| Input capacitance (output open-circuited) | 3240 | $V_{EB}$ = 0.5 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 8 | pF |
| Magnitude of small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 5 V dc; $I_C$ = 500 µA dc f = 30 MHz | $\lvert h_{FE} \rvert$1 | 1.0 | | |
| Magnitude of small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc f = 100 MHz | $\lvert h_{FE} \rvert$2 | 1.0 | 5 | |

See footnotes at end of table.



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* TABLE I. Group A inspection – Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 - Continued | | | | | | |
| Small-signal short- circuit forward current transfer ratio | 3206 | $V_{CE}$ = 10 V dc; $I_C$ = 1 mA dc<br>f = 1 kHz | $h_{fe}$ | | | |
| 2N3810, 2N3810L, U | | | | 150 | 600 | |
| 2N3811, 2N3811L, U | | | | 300 | 900 | |
| Small-signal short-circuit input impedance | 3201 | $V_{CE}$ = 10 V dc; $I_C$ = 1 mA dc;<br>f = 1 kHz | $h_{ie}$ | | | |
| 2N3810, 2N3810L, U | | | | 3 | 30 | kΩ |
| 2N3811, 2N3811L, U | | | | 3 | 40 | kΩ |
| Small-signal open-circuit output admittance | 3216 | $V_{CE}$ = 10 V dc; $I_C$ = 1 mA dc;<br>f = 1 kHz | $h_{oe}$ | 5 | 60 | μmhos |
| Small-signal open-circuit reverse-voltage transfer ratio | 3211 | $V_{CE}$ = 10 V dc; $I_C$ = 1 mA dc;<br>f = 1 kHz | $h_{re}$ | | $25x10^{-4}$ | |
| Noise figure | 3246 | $V_{CE}$ = 10 V dc; $I_C$ = 100 μA dc;<br>Rg = 3 kΩ, f = 100 Hz (see 4.5.8) | F1 | | | |
| 2N3810, 2N3810L, U | | | | | 7 | dB |
| 2N3811, 2N3811L, U | | | | | 4 | dB |
| Noise figure | 3246 | $V_{CE}$ = 10 V dc; $I_C$ = 100 μA dc;<br>Rg = 3 kΩ, f = 1 kHz (see 4.5.8) | F2 | | | |
| 2N3810, 2N3810L, U | | | | | 3 | dB |
| 2N3811, 2N3811L, U | | | | | 1.5 | dB |
| Noise figure | 3246 | $V_{CE}$ = 10 V dc; $I_C$ = 100 μA dc;<br>Rg = 3 kΩ, f = 10 kHz (see 4.5.8) | F3 | | | |
| 2N3810, 2N3810L, U | | | | | 2.5 | dB |
| 2N3811, 2N3811L, U | | | | | 2.0 | dB |
| Noise figure (wideband) | 3246 | $V_{CE}$ = 10 V dc; $I_C$ = 100 μA dc;<br>Rg = 3 kΩ, noise bandwidth 10 Hz to 15.7 kHz (see 4.5.9) | F1 | | | |
| 2N3810, 2N3810L, U | | | | | 3.5 | dB |
| 2N3811, 2N3811L, U | | | | | 2.5 | dB |
| Subgroup 5 and 6 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.
7/ When using table I, subgroup 2, as electrical end-points, this test is only required for JANS end-points, except subgroups C2 and C3.



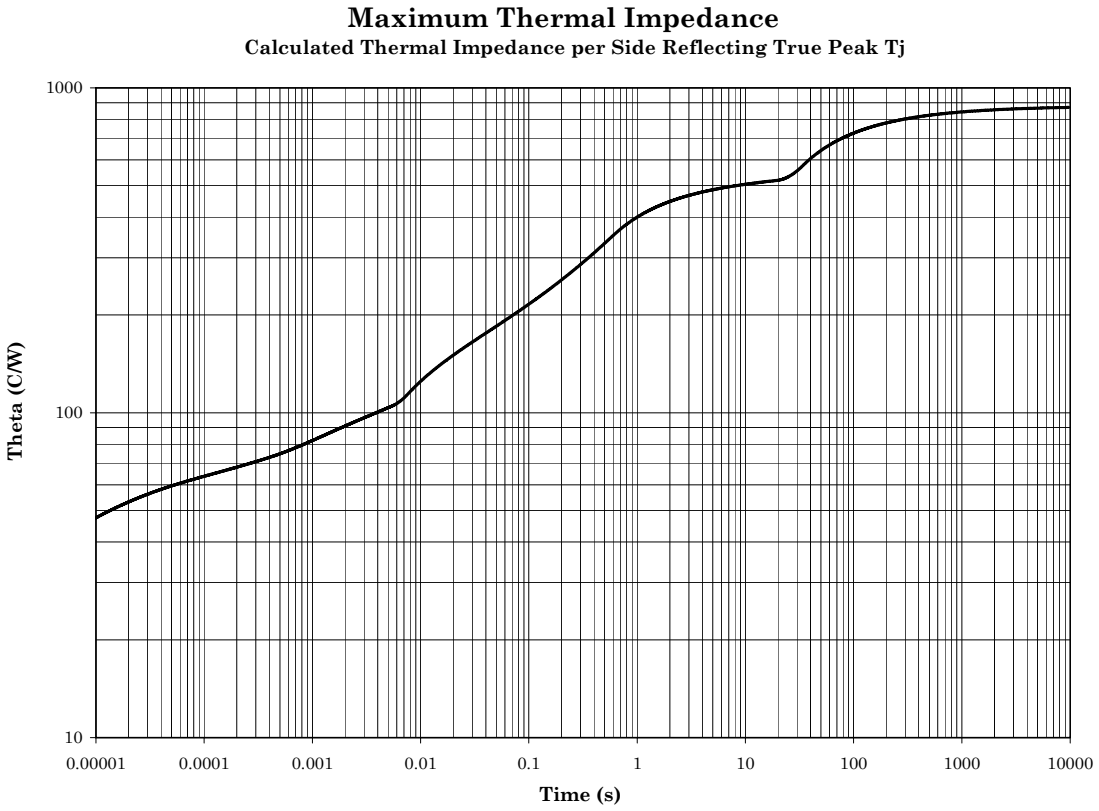
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* TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.3 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | Intermittent operation life: $V_{CB}$ = 10 V dc, 6,000 cycles | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.3 herein. | |
| Subgroup 4 | | | |
| Thermal response curves | | See 4.2.2.1. | Sample size<br>N/A |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices |
| Electrostatic discharge (ESD) | 1020 | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B | |



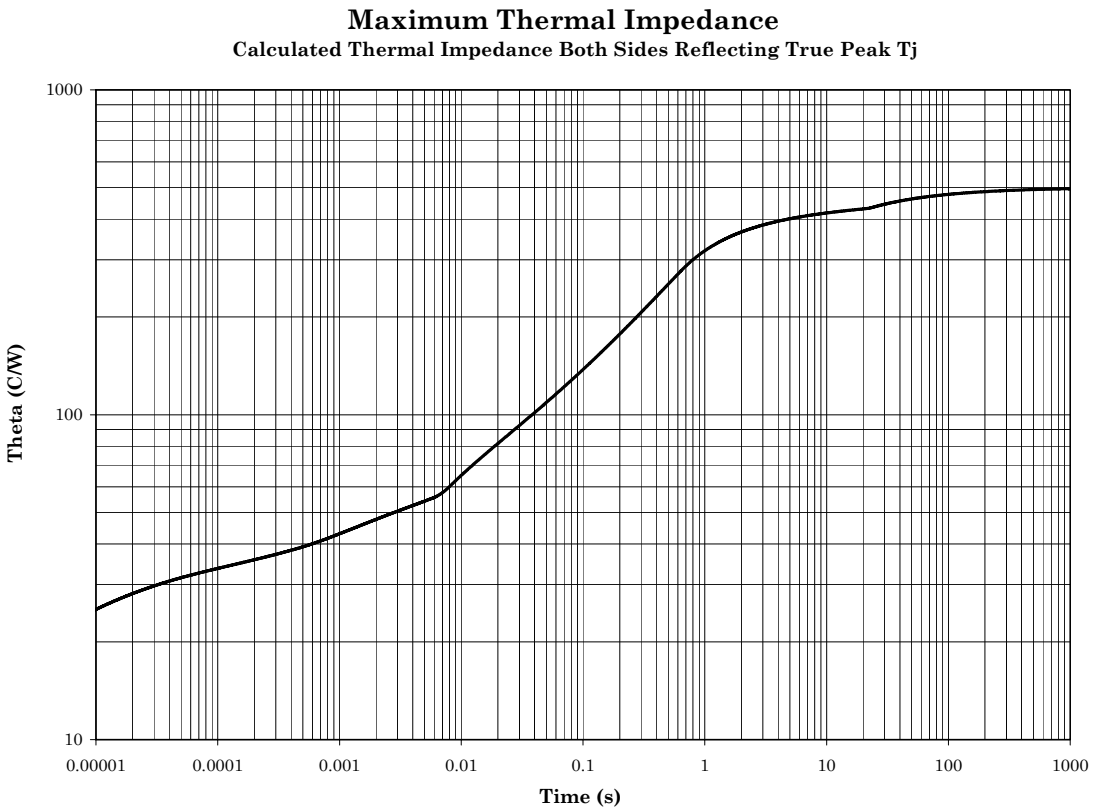
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* FIGURE 5. For each side: Thermal resistance = 875°C/W, Pt = 200 mW.



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* FIGURE 6. Both sides: Thermal resistance = 500°C/W, Pt = 350 mW.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3810) will be identified on the QML.

| Die ordering information | | |
|---|---|---|
| PIN | Manufacturer | |
| | 43611 | 34156 |
| 2N3810<br>2N3811 | JANHCA2N3810<br>JANKCA2N3811 | JANHCB2N3810<br>JANKCB2N3811 |



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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AV, MI
Navy - AS, MC
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

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